Decking Installation & Maintenance Guide

TimberTech Advanced PVC Decking

TimberTech®



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⚠ STOP! Read this section before you start!

Always make sure to visit TimberTech.com to ensure you are viewing the most current installation instructions, care and cleaning, technical information and more.

TimberTech Advanced PVC Decking, Fascia and Riser should be installed using the same good building principals used to install wood or composite decking and in accordance with the local building codes and the installation guidelines included below. AZEK® Building Products LLC and it's affiliates, successors, and assigns accept no liability or responsibility for the improper installation of this product. TimberTech Advanced PVC Decking, Fascia and Riser may not be suitable for every application and it is the sole responsibility of the installer to be sure that TimberTech Advanced PVC Decking, Fascia and Riser are fit for the intended use. Since all installations are unique, it is also the installer's responsibility to determine specific requirements for each TimberTech Advanced PVC Decking application. TimberTech recommends that all applications be reviewed by a licensed architect, engineer, or local building official before installation.

Tools Required

TimberTech Advanced PVC Decking, Fascia and Riser can be installed with a minimum number of readily available tools. All tools should be used per applicable manufacturers' instructions. Some of the basic tool requirements:

- Variable speed cordless drill
- Chalk Line
- Tape Measure
- Spacing Tools
- Square
- Miter Saw
- Safety Glasses
- Level
- String Lines
- Table Saw
- Jig Saw

Use the jigsaw to cut around obstructions such as posts. For best results, use a 10-12" sliding miter saw and always use with a clean, flat and sharp blade (12"-60 tooth minimum). With the top of the board facing up, start the cut with the miter saw/blade in the retracted position (furthest away from the operator). With the blade turning at full speed, lower the blade and pull the blade into the board toward you. Do not push the blade into the cut or cut from the top down unless using a non-sliding miter saw. If cutting with the board in the face-down position, start with the blade in the fully extended and lowered position and push the blade into the board to make cuts. When working with Advanced PVC products be sure to wear proper clothing and safety equipment. Safety glasses should be used during the entire installation process. Do not use any cordless saws.

STOP AND CALL TimberTech Customer Service at 1-877-275-2935 BEFORE starting the project if f you have an application not specifically addressed by this Installation Guide or when working on a large project.

Vinyl/Non-Porous Items/Rubber Back Mats

Do not use non-porous items such as rubber or vinyl products (i.e. welcome mats, planters, pool toys, tarps, etc.) on deck boards. A reaction can occur that causes discoloration of the decking under the rubber/vinyl product. This is a common caution for polymer decking products. Should discoloration occur, contact TimberTech Customer Service for assistance.

Static Electricity

The buildup or generation of static electricity is a naturally occurring phenomenon in many plastic based products such as carpeting, upholstery, and clothing, and can occur on alternative decking under certain environmental conditions. Static electricity is not a defect and can occur naturally. To temporarily relieve static build-up, spray a mixture of 1 part fabric softener to 20 parts water onto surface of deck. Reapplication may be needed periodically.

Color and Temperature

Although Advanced PVC Deck products are cooler to the touch than many other deck board products in similar colors, all decking products will get hot in the sun. Additionally, the darker the decking color, the hotter it will feel.

Natural Weathering

Depending on environmental conditions, TimberTech Advanced PVC Deck, Porch, Fascia, and Riser colors may appear to change over time as part of the natural weathering process consistent with the warranty guarantees where applicable.

Periodically move furniture, planters, rugs, etc. to facilitate more even weathering. If not, uneven weathering may occur and may appear different than areas with full sun exposure.

Additionally, decking installed under roof, awnings and shade from other structures may appear different than decking that is constantly exposed to direct sun.

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TimberTech Advanced PVC Decking

We recommend using fasteners that are approved for Advanced PVC decking. Some gapping will occur at board ends or butt joints. To help reduce gapping, use top-down fasteners such as TOPLoc® stainless steel or Cortex® and follow the installations tips and recommendations.

DO NOT use nails to fasten Advanced PVC Decking, Fascia and Riser boards.

DO NOT cut any metal product on or near Advanced PVC Decking (including aluminum, steel or other metal railing). The debris from the cut metal can become embedded in the deck surface which can damage the surface and/or rust.

Always be sure to check and cut the factory ends of boards to ensure they are square. If constructing deck with butt joints, consider shorter lengths such as 12' or 16' to help minimize gap size, or eliminate butt joints by using a divider board design. Be sure to properly frame for a divider board.

Walking Surface

Advanced PVC deck boards need to be installed embossed side up only.

Cantilevering/Extending Boards Past Framing

Do not extend or cantilever Advanced PVC deck boards more than ½" beyond the edge of any framing member or fascia.

Notching Deck Boards Around Posts/Obstructions

Always install framing/solid blocking below deck boards that have been cut or notched such as around posts or other obstructions to provide solid support for the deck boards in these locations. Do not fit deck board tight around posts, protrusions, or obstructions. Cut deck boards short leaving room for the boards to expand and contract against any solid structures including wood posts, metal, concrete, pavers, masonry, buildings, etc. Be sure to fasten notched deck boards to the framing/blocking below on both sides of the post using 2 top-down fasteners on each side.

Load Bearing

Advanced PVC Decking is NOT intended for use as columns, support posts, beams, joists, stringers, guard railing, or other primary load-bearing members. Do not construct walls, stone columns, roof supports, or other permanent load-bearing structures on top of TimberTech Decking products.

Routing

Advanced PVC Deck, Fascia and Riser SHOULD NOT BE ROUTED on the edges, ends, or sides with the exception of notching the start/stop boards as shown in this guide (under CONCEALoc® section).

Color and Grain Patterns

TimberTech Advanced PVC decking is designed with an authentic, natural wood visual. To create this visual appearance, we intentionally introduce varying degrees of variation in color or grain pattern in the surface of the boards. These variations may be more or less pronounced, depending on the decking collection and color selected (see dealer or TimberTech.com for more details). This is intentional and part of the manufacturing process, giving Advanced PVC Decking the most realistic and wood-like appearance possible. This variation is purely aesthetic and does not or will not affect the performance of the product.

When working with multiple units of decking, consider pulling boards randomly to mix rather than completing one unit at a time. It is always a good practice to lay out deck boards in different lighting conditions/angles before final fastening.

Air Flow Considerations

It is recommended that Advanced PVC decking not be installed directly over rubberized water management products that do not allow for installation of proper framing members and/or blocking. In areas with obstructed airflow such as roofs, water management systems, skirting, and on-grade applications, a minimum of 1- 1/2" clearance should be provided. Note: More clearance and airflow can help reduce/minimize movement and gapping that may occur as a result of elevated heat build.

Applications with obstructed airflow will experience elevated heat build which can lead to increased expansion and contraction/gapping. Therefore, consider face fastening with stainless steel screws or Cortex® to help hold deck boards and reduce gapping. When installed with hidden clip style fasteners and grooved deck boards, increased expansion and contraction/gapping may occur. Additionally, consider using shorter length deck boards and/or divider boards.

While Advanced PVC products are great for deck re-planking (removing old deck surface boards and installing Advanced PVC on code-compliant substructure), Advanced PVC Decking SHOULD NEVER be installed on top of an existing deck surface or deck boards or directly to concrete or masonry or floating sleeper systems. Do not install Advanced PVC Decking over any type of existing sub floor like plywood or sheathing type material such as OSB. TimberTech Advanced PVC Decking is designed to be installed over a standard footer/post/ beam/joist/sub-structure frame.

When installing Advanced PVC Decking in front of Low-E patio doors, sliders and windows that are within 30" or lower of deck surface, reduce joist spacing a minimum of 6' away from windows to 6-8" o.c. max and use top-down fastening.

Under rare conditions, with grooved decking installed on elevated decks that are prone to higher wind speeds (i.e. open areas, higher elevations, or coastal locations) wind generated noise such as howling, whistling, or groaning have occurred. If you plan to build an elevated deck in an area that is prone to more frequent and/or higher wind speeds, consider using square edge decking. If wind generated noise occurs with grooved decking, installing a gasket type material between edges of grooved deck boards, or installing a ceiling or wind baffling material to the underside of the deck structure, or enclosing the bottom of deck with a skirting or any method that reduces or eliminates wind from passing up and through into the spaced grooved edges of the deck boards will reduce or eliminate wind noise.

Failure to install TimberTech Advanced PVC Decking products in accordance with applicable building codes and this Installation Guide may lead to field performance concerns, aesthetic concerns, installation related issues, and personal injury, and affect deck system performance. Product failures caused by such failure are not covered by Timbertech's product warranty.

Floating Sub-Structures/Roof Top Decks

DO NOT attach deck boards to a floating sub-structure, which is a deck support system not mechanically fastened frequently, sufficiently, or permanently to post beams, concrete, roof top, building, or any other structural substrate. Never attach Advanced PVC Decking to sleeper systems, or between sleeper systems any loose framing members that are not properly blocked, braced, and securely fastened to footing or structure.

TimberTech Advanced PVC Decking is designed to be installed over a standard footer, post, beam, joist sub-structure frame. STOP AND CALL TimberTech Customer Service at 1-877-275-2935 for any other applications or help with large or custom projects.

When considering Advanced PVC Deck in a roof top application, it is imperative that the framing/sub-structure to which the decking will be attached is securely attached to the building or roof deck using suitable mechanical fasteners. Do not use adhesives to attach framing to the building or roof surface, and NEVER ATTACH DECK BOARDS TO A FLOATING FRAME/SUB-STRUCTURE.

Frequently check tops of sub-structure framing members to ensure that they are in plane, as Advanced PVC Decking will conform to the contour of the framing members to which it is attached. It may be necessary to shim or plane framing members due to thickness variations in lumber, or variations in the surface to which the framing/sub-structure is attached. Securely attaching rigid blocking between framing members at frequent intervals should also be incorporated to help limit sub-structure movement. Framing or sleepers must be secured and structurally sufficient to resist lifting of frame or sleepers, especially on board ends, and resist thermal movement of decking in any direction, or an uneven walking surface may appear. Insufficient framing or sleepers will also demonstrate increased gapping at butt joints and miters of your Advanced PVC deck board.

Proper drainage and air flow should also be considered, with a minimum of 1 ½" sleeper height necessary and deck boards must be attached to the properly secured substructure with a minimum of a 1/8" gap between deck boards and ensure that sleepers are securely attached to building or sub-structure—NOT FLOATING.

It is the installer's responsibility to determine if Advanced PVC Deck is suitable for all applications, including roof-top applications. TimberTech recommends that all applications be reviewed by a licensed architect, engineer, or local building official before installation. If you have any questions or need further assistance, please call TimberTech Customer Service at 1-877-275-2935 or visit our website at TimberTech.com.

Extreme Heat Warning

Be aware of excessive heat on the surface of TimberTech Advanced PVC products from external sources, such as but not limited to, fire, hot ashes/embers, fire pits, grills or reflection of sunlight from energy-efficient window products. Reflection of sunlight from Low-emissivity (Low-E) glass can potentially harm TimberTech Advanced PVC products. Low-E glass is designed to prevent passive heat gain within a structure and can cause unusual heat build-up on exterior surfaces. This extreme elevation of surface temperatures, which exceeds that of normal exposure, can possibly cause TimberTech Advanced PVC products to melt, sag, warp, discolor, increase expansion/contraction, and accelerate weathering. You must add additional framing, tighter joist spacing (6 to 8" maximum) in areas with Low-E reflectivity or exposure. Also, only use top-down fastening in these locations such as near glass, areas of high heat build or sunlight reflection. Current or potential TimberTech customers that have concerns about possible damage by Low-E glass should contact the manufacturer of the product which contains Low-E glass for a solution to reduce or eliminate the effects of reflected sunlight.

Metamerism

Metamerism is the phenomenon wherein two colored samples will appear to be identical under one light source or viewing angle, but will appear to be different shades under a 2nd source or viewing angle. Metamerism can occur even when the deck boards are cut from the same deck board or come from the same unit. This phenomenon can occur when such boards are installed in opposing/different directions.

When installing Advanced PVC deck boards at different angles or different directions, metamerism may make the boards installed at one angle appear to be of a different shade than the deck boards installed at the other angle, depending on the lighting and the angle which the deck is viewed. This is a natural phenomenon, not a defect. Metamerism is an aesthetic phenomenon and not a performance or product warranty issue.

⚠ Excessive Construction Debris

It is important during construction that the deck's surface stay clear of excessive build-up of dirt, sand, and dust. Do not allow dust or debris from tile, concrete, landscape blocks, or other masonry products to settle on the surface of the deck. If these materials are not removed immediately, the deck surface will become difficult to clean and can potentially damage the deck's surface finish.

If a build-up does occur please refer to the Care and Cleaning section in the Advanced PVC installation guide, the TimberTech Decking Care & Cleaning Guide, or our website, TimberTech.com. Do not use Advanced PVC Deck as a work surface.

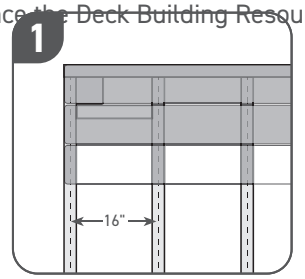
When installing Advanced PVC Deck, Fascia, and Riser, best practice is to install after all other construction (siding, roofing, painting, etc.) is completed, otherwise permanent damage may occur.

IMPORTANT: Do not allow airborne dust from concrete, landscape blocks, or any masonry product, including fiber-cement siding and trim, to accumulate on the Advanced PVC Deck, Fascia, and Riser surface as it may damage the surface of the deck.

Substructure/Preparation

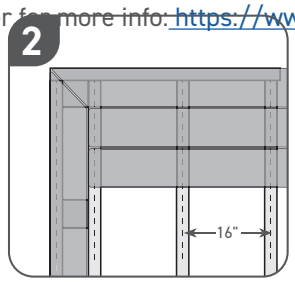
Follow these guidelines for deck installation:

- Framing/sub-structure must be securely attached to ground footings or building structure. **No floating sub-structures** should be used in any application with Advanced PVC Deck, including on grade, concrete, or in roof-top applications.
- For aluminum substructure please go to https://www.timbertech.com/products/aluminum-framing/
- As joists are installed, run string line frequently to ensure that the tops of the joists are in plane, as Advanced PVC Decking will conform to the contour of the joists to which it is attached. It may be necessary to shim or plane joists due to size variations in lumber. Be sure to sort and exclude any extremely crowned joists/lumber material.
- Prior to installation, check to make sure all joists are level, structurally sound, and that there are no nails or screws protruding.
- Proper joist spacing is required for proper installation. Joist spacing should never exceed 16" on center (up to 24" on center for TimberTech MAX boards). For a more rigid feel, less deflection and additional fastener hold, consider 12" on center for best results. For all angled or commercial installations of decking install joists 12" on center maximum.
- If installing decking over existing framing, ensure that all joists/framing members are structurally sound, free from rot/decay, and tops of framing are in plane across tops. Replace or add new framing members as needed to ensure structure is sound and all framing members are in plane across tops.
- **You MUST install SOLID WOOD BLOCKING** between each joist, placed in rows at maximum every 4' - 6' within the structure using exterior grade screws. Solid blocking will help reduce up or down movement and/or twisting of joists. If twisting up or down movement of joists occur, Advanced PVC Decking will follow the contour of the joists and appear wavy. Limiting joist movement can help reduce excessive gapping and also ensure a more uniform finished surface of the Advanced PVC Deck.
- For better results use TimberTech PRO-Tac joist tape to provide protection for wood joists and framing.
- **Use additional framing/blocking, tighter joist spacing (6 to 8" maximum) and only top-down fasteners (i.e. TOPLoc® or Cortex® for Advanced PVC) in areas of high heat build or sunlight reflection such as locations near glass.**
- Reference the Deck Building Resource Center for more info: https://www.timbertech.com/resources/deck-building



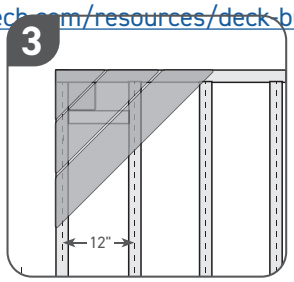
16" BASIC INSTALLATION SUBSTRUCTURE

16" on center maximum joist spacing (up to 24" on center for MAX boards), (12" on center for commercial applications)



PICTURE FRAME INSTALLATION SUBSTRUCTURE

16" on center maximum joist spacing (up to 24" on center for MAX boards) plus additional support for picture frame structure (12" on center for commercial applications)



DIAGONAL INSTALLATION SUBSTRUCTURE

12" on center maximum joist spacing (up to 16" on center for MAX boards). More severe angles may require closer joist spacing.

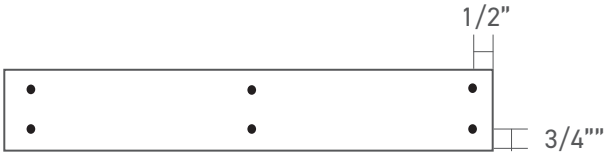
Additional diagrams can also be seen on https://www.timbertech.com/resources/deck-building/deck-joist-spacing/.

Expansion and Contraction

Similar to many building products Advanced PVC Deck/Fascia/Riser will expand and contract, move, and gap with changes in temperature, unlike wood whose size will fluctuate with moisture changes. The expansion and contraction encountered may result in gaps, which might be noticed at the ends of boards or in splice/butt joints. Gapping is normal and some gapping can be expected. Using our recommended top-down fasteners to install the deck boards according to the Installation Guidelines will help minimize expansion and contraction.

Note: Splices/butt joints are where gapping most commonly occurs. Limiting length of deck boards for long runs can help minimize gapping. Adding design features, such as properly framed parting or divider board, can reduce or eliminate the need for splice, miter, or butt joints. Also consider using shorter deck board lengths or a design that allows running deck boards in shortest possible direction. Properly frame divider boards using cross blocking and ladder blocking with proper screw attachment (do not use nails). Click here to watch divider board framing and install best practices

- **Keep decking as cool as possible during attachment and use top down fasteners.** Direct full sun exposure will increase board temperature and length. If attaching deck board during high heat/full sun exposure, deck board will increase in length. If deck board is attached when longer length has occurred, upon cooling, larger gapping can be expected.
- Install PVC deck boards ends tight together. **DO NOT leave a gap at butt joints, splices or miters, or where deck boards meet with parting, breaker or dividers boards, and picture-framing boards regardless of temperature when installing PVC boards end to end or end to side.**
- Expansion/contraction is most significant where extreme temperature change may exist. Expect larger gaps at butt joints/miters when PVC decking is attached at elevated temperatures or higher heat build conditions.
- Cut and fasten the deck boards as soon after cutting as possible.
- Be sure to fasten ends of boards, splices, or abutments to building or structure securely into framing using a minimum of 2 screws. Position the screws at or within 1/2" of each board end or joint and 3/4" from side of deck plank. Failure to do so could lead to lifting of deck board ends and increased gapping.
- For butt joints, be sure to trim cut factory ends to ensure squareness. Do not use factory end cuts. **Do not scarf cut butt joints.**
- Do not fit deck board tight around posts, protrusions, or obstructions. Cut deck boards short, leaving room for the boards to expand and contract against any solid structures including metal, concrete, pavers, masonry, buildings, etc. When cutting and fastening Advanced PVC deck boards against Advanced PVC deck boards, do not leave gaps at divider boards, butt joints, or mitered corners.

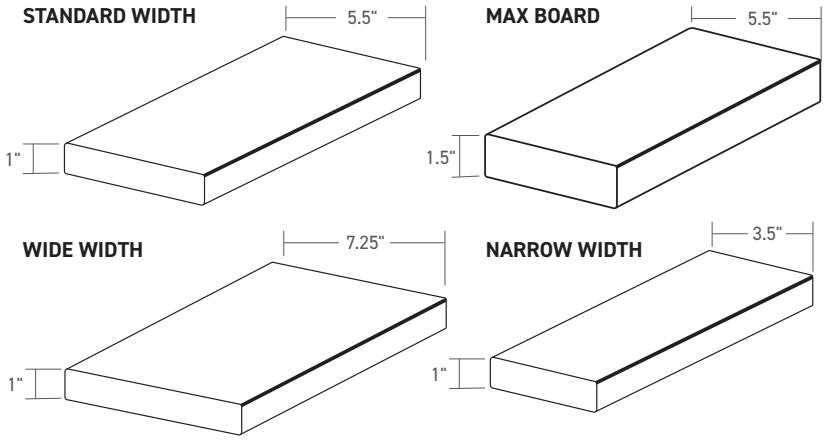


- To help reduce or minimize movement, gapping, and expansion/contraction with Timbertech Advanced PVC Decking, consider using only top-down fastening methods such as Cortex® or TOPLoc® screws for the entire deck while avoiding high heat build conditions at time of attachment as was described above. Refer to "Cortex® Hidden Fastener Installation- For TimberTech Advanced PVC" and "TOPLoc® Fasteners Installation- For TimberTech Advanced PVC" videos for more details https://www.timbertech.com/products/fasteners/
- For best results and a better hold, install 3 fasteners within 1/2" of all board ends.
- Provide a 1/8" to 1/4" side to side gap between deck boards.
- You must stagger splices and miters across deck surface so that all splices and miters do not break on the same joist or beam. Consider using divider boards with properly constructed framing to eliminate butt joints. See Divider board design.
- To further minimize expansion and contraction, proper ventilation and air flow should be considered.
- Some area building codes may require double joisting when joining boards end to end. Double joist and/or secure double blocking is always required at butt joints when using CONCEALoc® fasteners. Always check with local code authorities for specific requirements before installation.
- During installation, frequently check that Advanced PVC deck boards are installed straight and with consistent gapping and spacing by using a string line.
- Other considerations: Darker colors promote higher heat build which can lead to larger gapping at board ends/butt joints.

Fasteners

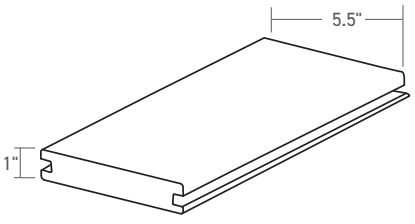
- No predrilling is required with Advanced PVC Decking.
- Due to the durability, longevity, and other unique characteristics of TimberTech Advanced PVC decking, TimberTech strongly recommends the use of a top-down fastening methods using only one of the following fasteners: Cortex® hidden fasteners, Advanced PVC TOPLoc® stainless steel color-matched deck, fascia and Riser fasteners, or a high-quality stainless steel screw. (minimum #7 size and minimum length of 2-1/4") (2-3/4" for TimberTech MAX boards)
- Be sure that screws are driven to a minimum depth of 1 1/4" into solid wood framing below the bottom of the Advanced PVC deck board. This will enhance lateral holding power and uplift resistance.
- Avoid using flathead screws. **As always, any fastener should be tested on a sample board before being used on decking.**
- Be sure to install a minimum of 2 screws per joist on each deck board. Regardless of temperature during attachment of Advanced PVC deck boards, be sure to keep butt joints and miters tight and place fasteners within 1/2" of ends. (Target 3/8" from ends). Drive all screws perpendicular into joist, never drive fasteners on an angle. If you cannot drive perpendicular, consider adding a sister joist or blocking.
- Use CONCEALoc®® hidden fasteners consider using three top-down fasteners within 1/2" of ends of deck boards (miters, butt joints, ends of boards at board ends.)
- The fastening pattern and installation methods are the same for standard width (5.5"), narrow width (3.5"), wide width (7.25"), and MAX (1.5" thick) boards. For best results, consider using three top-down fasteners at every joist as well as board ends when installing Advanced PVC wide width (7 1/4" deck boards).For salt water coastal applications, we suggest using the above minimum fastener requirements in 316 stainless steel.
- For TimberTech Advanced PVC grooved decking, use CONCEALoc®® hidden fasteners.
- The following fasteners are compatible with MAX 1.5" thick boards:
    - 2-1/2" Cortex® for MAX (305 stainless steel)—for coastal applications use Cortex® (316 Stainless Screws)
    - 3" TOPLoc® for TimberTech Composite - Light Gray (305 stainless steel)
- TimberTech recommends using TimberTech Fasteners with TimberTech Decking. TimberTech Fasteners are specifically engineered to deliver the best results and to minimize the risk of issues during or after installation, including but not limited to gapping, warping, board movement, oil canning, and squeaking. These types of issues are not covered by TimberTech's product warranty and TimberTech will not be financially liable in the event such issues arise.
- Use a scrap of the decking material or tapping block made of material that is softer than the deck boards and only use the minimum amount of force necessary to position the deck board during installation of fasteners.
- WARNING: When installing TimberTech Advanced PVC with any fasteners, always use a beater board against side of deck board. Do not strike deck board directly with hammer or rubber mallet or damage to the deck plank may occur.
- WARNING: When installing fasteners, or working on the deck surface, some shoe soles or harder knee pads may leave scuffs when dragged across deck surface. Shoe covers are recommended.

Square-Shouldered Deck Boards



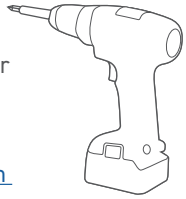
Grooved Deck Boards

Use TimberTech CONCEALoc® Hidden Fasteners. Follow the directions on the CONCEALoc® bucket, box or pages 8-11 of this guide.



Cortex® Hidden Fastening System Installation Instructions

Suggested Drills

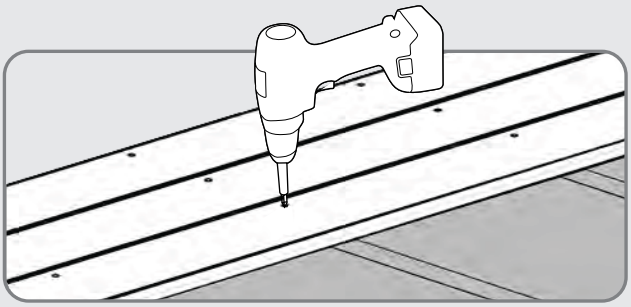


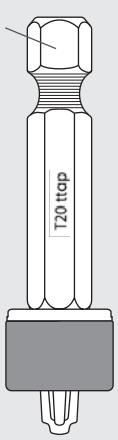
Corded or cordless drill or impact driver (18V variable speed recommended).

Click here to watch: How to Install TimberTech Advanced PVC Decking with Cortex® Hidden Fasteners.

1 DRIVE SCREW

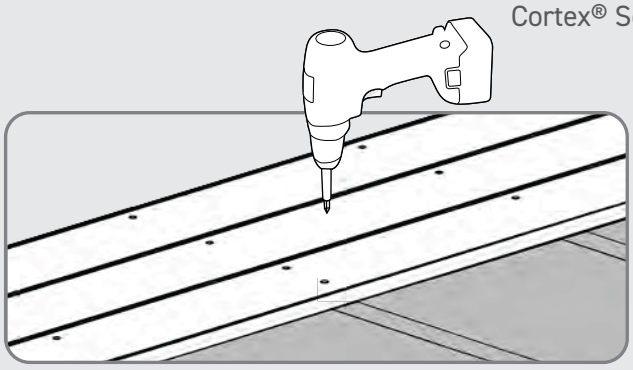
- Using setting tool provided, drive screw at medium or high speed until it disengages at the proper level below the deck surface.
- Drive at one continuous speed, do not slow down or toggle the drill trigger in a start/stop manner.





2 INSTALL DECK SCREWS INTO JOISTS

- Install two screws into decking at each joist. For best results, use 3 top down fasteners at miters, butt joints, and board ends.
- Maximum spacing between each row of screws should be 16" on center (up to 24" on center for MAX boards). For a more rigid feel, less deflection, and additional fastener hold, consider 12" on center for best results.
- Reference Pages 4, 5 & 6 for proper fastener placement and allowable spans for deck boards.

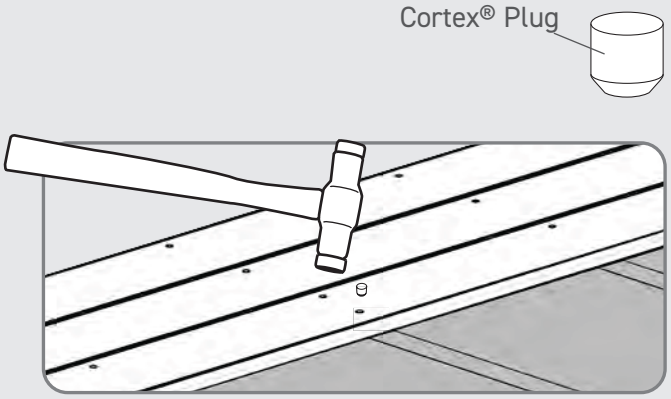




Note: For all angled or commercial installations of decking, install joists 12" on center maximum.

3 INSTALL PLUG

- Place plug into cored hole with grain aligned and gently tap flush with deck surface.
- Always use a smooth-faced plastic hammer to avoid damaging the deck surface while setting the Cortex® plug.
- When installing Collated Cortex® Plugs:
    1. Tear off one strip of collated Cortex® plugs
    2. Position the leading plug into hole with grain aligned
    3. Gently tap plug into place
    4. Pull collated strip away from installed plug
    5. Tap plug once more until flush



IMPORTANT: When installing Cortex® plugs, use caution to prevent damage/ denting to the Advanced PVC deck.

Important Information

- The Cortex® Setting Tool must be used to set the Cortex® screw to the proper depth.
- The Cortex® fastener must be driven perpendicular to the deck surface for proper fit and ideal finished look (DO NOT drive on angle).
- If the butt joint does not break evenly on a joist, add a sister block/joist to ensure screw is driven perpendicular to surface.
- The cored hole must be free of debris or moisture. Use a smooth plastic hammer head to set the Cortex® plug.
- Issues arising from use of non-Cortex® Screws or plugs are not covered by TimberTech's product warranty.
- Fasteners should be positioned within 1/2" from board ends and 3/4" from board sides.

Restricted Cortex® Uses

The Advanced PVC Cortex® System is designed for traditional elevated deck surfaces. The following applications require a different fastening method:

- Advanced PVC Cortex® should not be used with Advanced PVC Fascia or any other material that is 5/8" thick or less. Cortex® fasteners may be used with TimberTech Advanced PVC Riser boards as they are 3/4" thick.
- Docks or decks with framing members in constant contact with saltwater.
- Standard Cortex® cannot be used with framing less than 2" thick due to screw length being too long. Standard Cortex® cannot be used with metal framing; for all TimberTech Aluminum Framing, use Cortex® Fasteners.

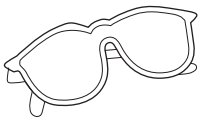
CONCEALoc® Hidden Fasteners for Use with TimberTech Advanced PVC Deck Boards

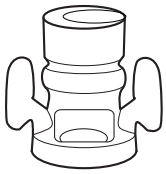
CONCEALoc® & Accessory Installation Instructions

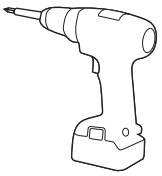
- Note: When using clips with Advanced PVC, larger gaps can be expected at butt joints or breaker boards. Consider using top-down fastening methods such as Cortex® or Advanced PVC TOPLoc® stainless steel color matched screws to help reduce gapping.
- CONCEALoc® Hidden Fasteners – retail box for 100-square-foot coverage or bulk bucket for 1,000-square-foot coverage. Includes stainless steel screws.
- CONCEALoc® Gun Pail – 500-square-foot coverage for use with the TigerClaw pneumatic installation tool sold by FastenMaster. Includes quality coated pneumatic fasteners. Stainless steel fasteners for coastal applications available through FastenMaster. Pneumatic tool must be tuned specifically to install lightweight Advanced PVC Decking. Failure to do so can result in less than desired securing/tightness of fastener, boards lifting from joists, boards feeling loose, or movement and could generate squeaking/noise. Double check to ensure deck board and clips are tight to joist; also be sure that clips are tight to deck board. You may need to hand tighten any loose clips that were installed with TigerClaw pneumatic tool
- For a hidden fastener surface when using CONCEALoc®, use Cortex® on the first and last boards.
- CONCEALoc® Router Bit – Should only be used for starter boards, picture frame boards, or other small areas if necessary— do not use this technique to install an entire deck. Rout the correct notch on a solid plank for a suitable CONCEALoc® fastener fit, which also allows for notching an installed plank over a joist.
- Reference the Deck Building Resource Center for additional guides: https://www.timbertech.com/resources/deck-building

CONCEALoc® Hidden Fasteners Installation Instructions

Tools Needed:


Safety Glasses


Router with Slot Cutting Bit (Optional)

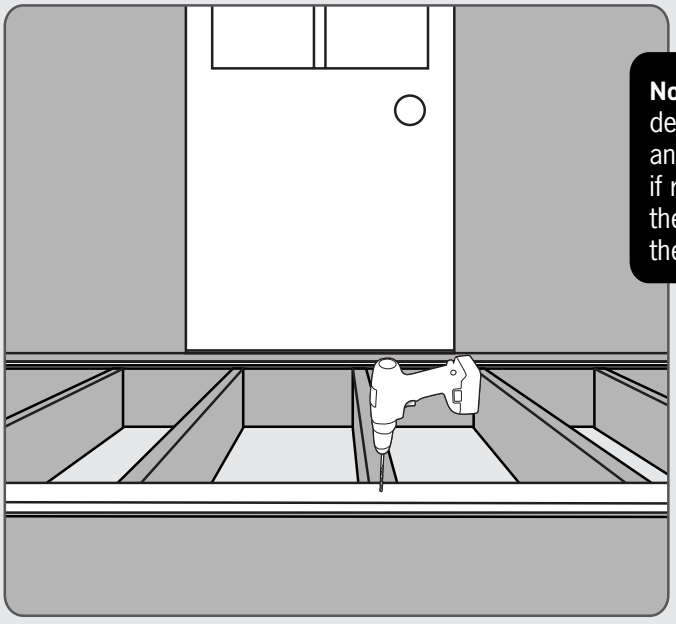

Drill


CONCEALoc® Screw & Clip

1

INSTALL THE FIRST PLANK

- Fasten outer edge of the first plank at each framing support 3/4" from the edge using Advanced PVC TOPLoc® or Cortex® screws. For a hidden fastener surface, use Cortex® on the first and last boards.

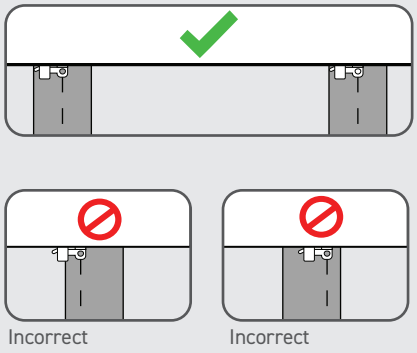


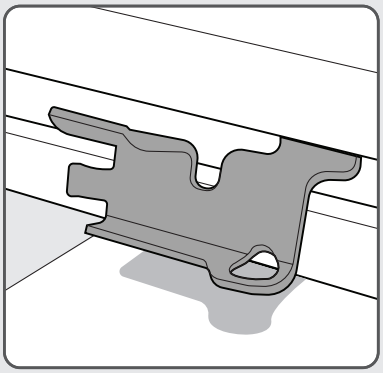
Note: Consider starting first deck board on outside of deck and work toward house so, if rip cut boards are needed, they will be concealed against the house.

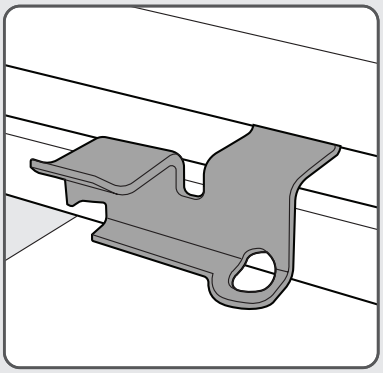
2

POSITION CONCEALoc® FASTENERS

- Fully insert the CONCEALoc® fastener into grooved edge of plank. Screw hole should be lined up with the center of support joist. Clip should be tilted upward slightly to fully insert into the groove/slot.

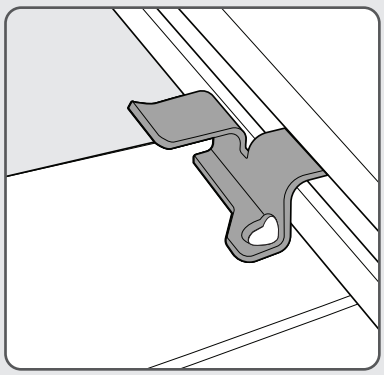


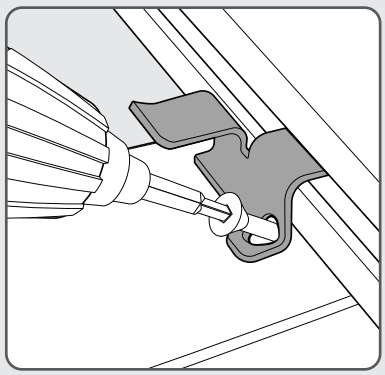


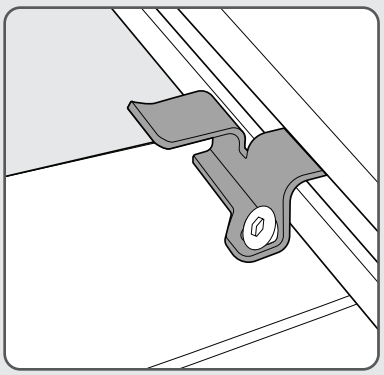


3

- Install CONCEALoc® screw at 45° angle through the fastener and into the joist while applying pressure on the fastener.
- Install one fastener and screw at each joist location. See specific and additional requirements for butt joints.
- Make sure fastener is in full contact with edge of plank and the screw is holding the fastener down tight to the deck board.

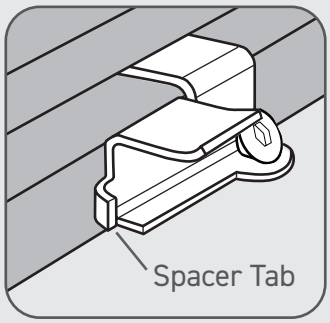


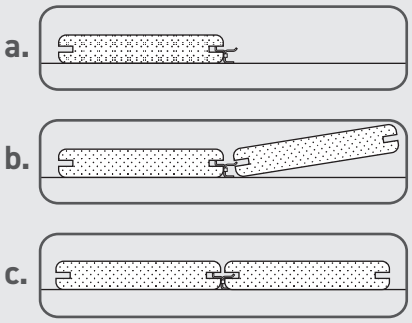


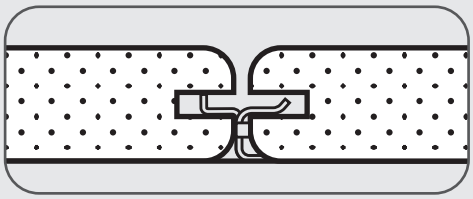


4 COMPLETE INSTALLATION

- Place the next plank into position against the fasteners. Slightly raise the outer edge of the plank being installed and slide it onto the fastener until the plank contacts the spacer tab. When properly installed the inserted plank should be in contact with the bent-forward spacer tab at the left end of the clip. To achieve this, use wood 2x4 at least 24" in length and use it as a tamping block/beater board to bring edge of deck board into contact with spacer tab to ensure proper spacing. **Important: Do not overdrive or bending of clip may occur.**
- Boards should be checked frequently for straightness, as they are installed using a string line or laser line.
  Never kick or strike the edge of the decking directly with a hammer or any object. Always use a beater board to disperse impact over a large area or damage may occur.





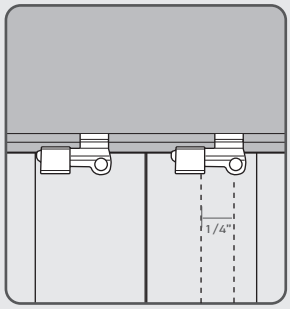


Correct placement

5 INSTALLING CONCEALoc® FASTENERS AT BUTT JOINTS

- **Joists at butt joints must be doubled up for proper fastener attachment.**
- Install fasteners as detailed below. Note that the two clips on the left are shifted approximately 1/4" to the right of center on the joists. If this is not done the spacer tab may fall within the gap of the butt joist (Fig 1).
- For additional hold and to help minimize gaps at butt joints, consider installing Cortex® screws within 1/2" of board ends as well as 2 screws per joist at a minimum of 3 joists from butt joints on both ends of the deck board.
- Never place Cortex® fasteners only/solely on a single joist for butt joints, board ends and miters. If using Cortex® with CONCEALoc®, you must place Cortex® into a minimum of 3 joists from each end of the deck board.

Note: Adding more Cortex® fasteners will help reduce gapping and increase performance. In addition to statement above, consider adding additional Cortex® fasteners for longer runs or longer length boards.



Joist 1 Joist 2

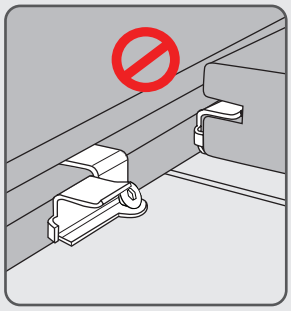
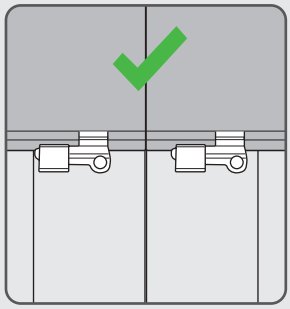
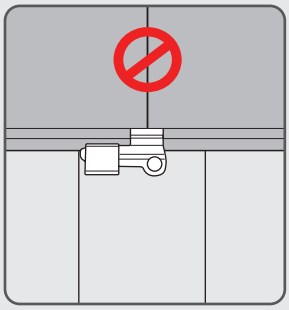


Fig.1



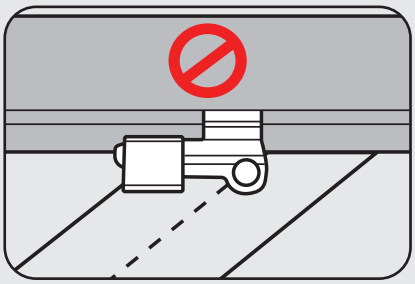
Joist 1 Joist 2



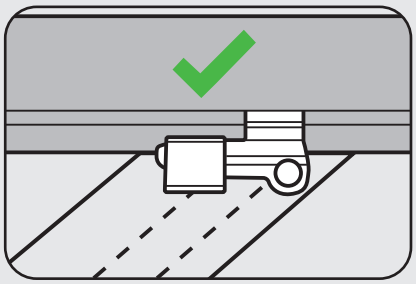
6

HOW TO INSTALL ON DIAGONAL

- When the deck is located in a corner of the house, start with a small triangular piece of decking on one corner and work away until last triangular decking piece is installed.
- Offset the clip 1/2" to the right of center line of the joist so screw will not exit the side of the joist when driven.



Incorrect



Correct: Shift over 1/2"

Note: Small triangular pieces of deck should be secured with Cortex® fasteners.

7

HOW TO REPLACE A PLANK

- Make two parallel cuts down the center of the board to be replaced, then remove the center piece.
- Using a small pry bar, remove the remaining pieces of decking from the tabs.
- Use a table saw to remove the bottom groove leg from one side of the replacement board, or mark board where clips will be located and notch the bottom leg of the groove where the clips will be located using a router or oscillating tool.
- Position new board into place and drive onto remaining tabs.
- Once the board is set into position, secure the loose edges with Cortex® hidden fasteners or TOPLoc® stainless steel fasteners.

8

TO USE CONCEALoc® HIDDEN FASTENERS WITH ADVANCED PVC SOLID DECK BOARDS (FOR START/STOP ONLY)

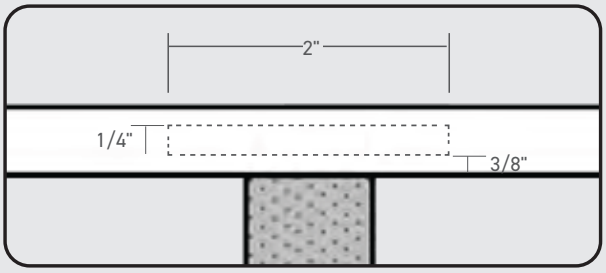
⚠ DO not use CONCEALoc® Hidden Fasteners for stair installation. Use solid square edge deck boards with Cortex®.

- Use Advanced PVC CONCEALoc® Groove Cutting Router Bit or a Freud model #56-112 or Bosch model #85610M 1/4" slot cutting bit.

  Tip: One of the benefits of the CONCEALoc® Router Bit is that it will pass over the joist. There's no need to flip the board or move it.
- Flip the plank over and start the notch 3/8" off the bottom of the plank.
- Set the depth of the cut from 7/16" to 1/2" deep.
- Create a 2" long notch on the edge of the plank at every joist. Do not rout/groove full length of deck plank. Only notch 2" in width above each joist to accept fastener.

TO USE CONCEALoc® HIDDEN FASTENERS WITH ADVANCED PVC SOLID DECK BOARDS (FOR START/STOP ONLY)

- Tip: Using a grooved board to locate the position of the router bit saves time and ensures consistency with notch location.



SIDELoc™ Hidden Fasteners for Use with TimberTech Advanced PVC Deck Boards

IMPORTANT: The SIDELoc™ Hidden Fastener System is the key to installing a beautiful, fastener-free surface on capped polymer decking. Designed specifically for Advanced PVC square-shouldered deck boards, SIDELoc™ Capped Polymer Deck Fasteners (referred to as SIDELoc™ Screws), SIDELoc™ Guide, and SIDELoc™ Driver Bits are all a part of the SIDELoc™ System. Each product was engineered to work exclusively within the system. Using tools, guides, fasteners, or bits from other brands will impair performance and won't produce the desired results.

- Consider using top-down fastening methods such as Cortex® or Advanced PVC TOPLoc® stainless steel color-matched screws to help reduce gapping at board ends and butt joints.

NOTE: SIDELoc™ should NOT be used with TimberTech Composite Decking- or any grooved profiles of decking. For best results with 7 1/4" decking, consider using top-down fasteners such as Cortex® into a minimum of 3 joists from each end of deck board in addition to SIDELoc™.

- For best results and a better hold, install top down fasteners such as Cortex® into a minimum of 3 joists from each end of the deck board in addition to SIDELoc™.
- Joists at butt joints must be doubled up for proper fastener placement.
- For picture frame miters, Cortex® hidden fasteners should be used. Install 3 fasteners along each edge of the miter joint.

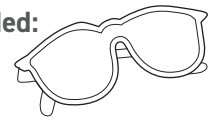
The SIDELoc™ Guide directs SIDELoc™ Screws into the side of the deck board and into the joist. Use SIDELoc™ Guides on any 3 1/2"–7 1/4" (89mm–184mm) wide solid capped polymer deck board. No predrilling is required, and the SIDELoc™ Guides provide automatic 1/8" (3mm) spacing between deck boards.

SIDELoc™ & Accessory Installation Instructions

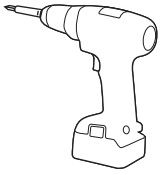
- SIDELoc™ Screws – available in Fastener Packs of 350, 700, and 1750
- SIDELoc™ Guide – SIDELoc™ 5 1/2" or SIDELoc™ 3 1/2"
- For wide-width boards, use the SIDELoc™ 7 1/4" Extension
- SIDELoc™ Driver Bit (included with SIDELoc™ Screws)
  - Driver Bits set the depth of drive making it impossible to overdrive the screw
- Drill – for maximum driving efficiency, set the drill speed to the highest drill setting

SIDELoc™ Hidden Fasteners Installation Instructions

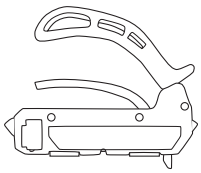
Tools Needed:



Safety Glasses



Drill



SIDELoc™ Guide for Correct Board Width



SIDELoc™ Screws

1

SET UP SIDELOC TOOL

The steps noted below must be followed to properly adjust the SIDELoc™ Tool for the varying requirements of the profiles used in Advanced PVC Decking.

- Adjusting the SIDELoc™ Tool: Loosen the thumb dial to adjust the spacer teeth to fit the width of the deck boards you are using. Once you have achieved a snug fit on the board, tighten the thumb dial to hold the position.
- Installing wide boards? Attach the SIDELoc™ 7 1/4" extension to the SIDELoc™ 5 1/2" guide to easily fasten 7 1/4" boards. For best results with 7 1/4" decking, consider using top-down fasteners such as Cortex® into a minimum of 3 joists from each end of deck board in addition to SIDELoc™.

2 HOW TO INSTALL ON ANGLE

- Position the SIDELoc™ Guide over the edge of the high side of the joist so the screw is fastened completely into the joist when fully driven.
- For precise positioning, draw a line using a straight edge or edge of deck board where the board intersects with the joist. Position the SIDELoc™ Guide so the middle of the screw guide rests over the line, then drive the screw. Repeat this process down the length of the deck board on both edges.

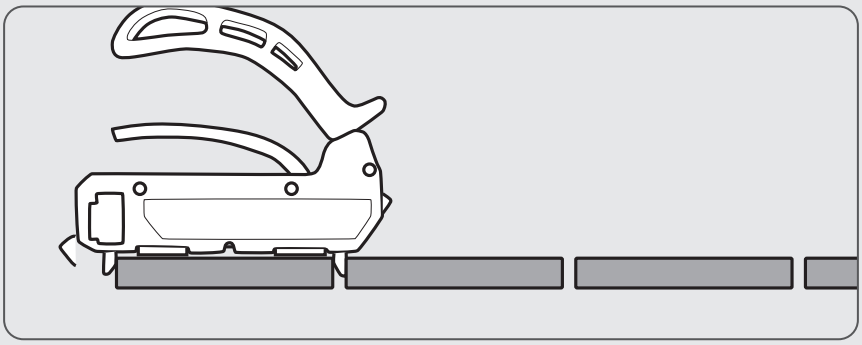
3 FASTENING THE BOARD NEXT TO THE STRUCTURE

- If the board next to the structure runs parallel to the structure, it may be necessary to face-fasten the inside edge of that board:

1. Place the board snug to the structure and mark where you'll drive the SIDELoc™ Screws into the joist.
2. Move the board away from the structure, position the SIDELoc™ Guide over the mark, and drive the SIDELoc™ screw half an inch (12.7mm) into the board. Repeat at every mark.
3. Push the starter board up to the structure with the SIDELoc™ screws on the outside edge of the board and drive the SIDELoc™ screws the rest of the way. Face fasten the inside edge with Cortex® or TOPLoc® for Advanced PVC.

4 SETTING THE SIDELOC GUIDE

- Beginning on one end of the deck board, squeeze the trigger handle of the SIDELoc™ guide and set the guide over the deck board, centered over the joist.
- With the tool feet flat on the surface of the board, release the trigger handle, allowing the spacer teeth to grip the deck board. A properly adjusted guide clamps securely to the board.
- With the SIDELoc™ guide in place on the deck board, push the deck board snugly against the starter board. The spacer teeth automatically provide a 1/8" (3mm) gap between the boards.

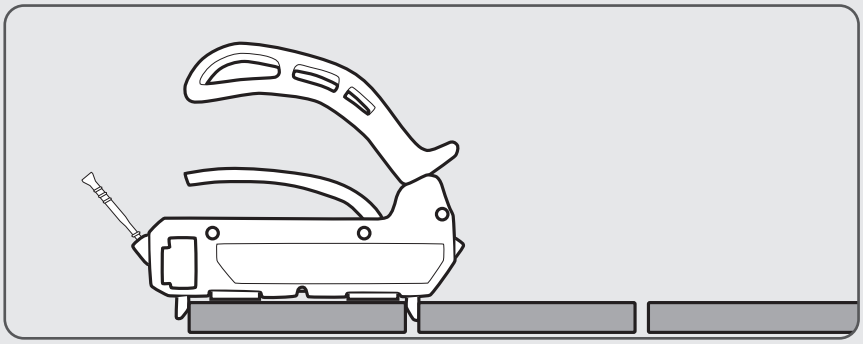


Note: Joists at butt joints must be doubled up for proper fastener attachment.

SIDELoc™ Hidden Fasteners Installation Instructions

5 LOAD

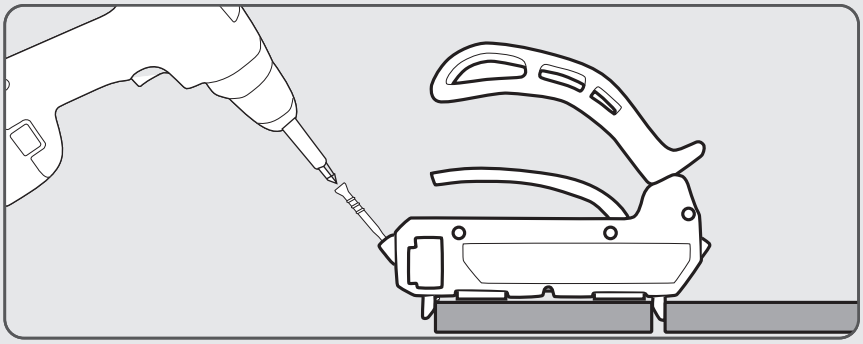
- With the SIDELoc™ Guide set on the board, place a SIDELoc™ Screw into each of the screw guides.



To ensure proper performance of the SIDELoc™ Guide, it is important to regularly check the screw guides for deck material build-up. If material is present, use the SIDELoc™ Driver Bit to clear the screw guides.

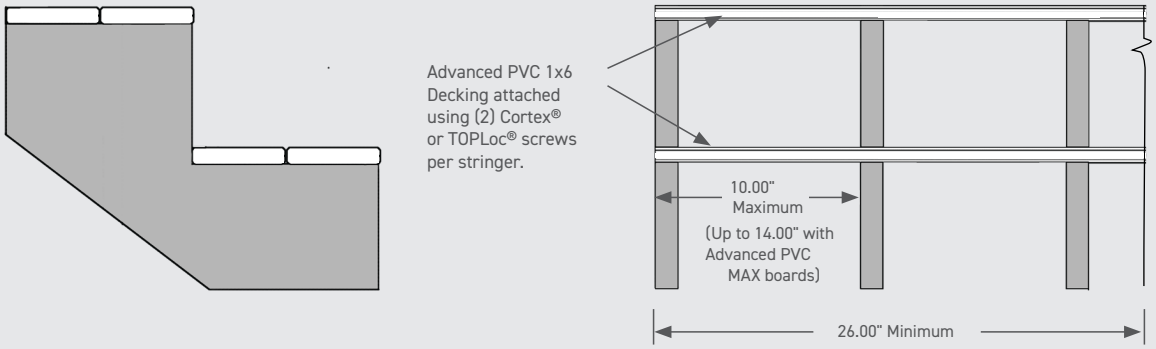
6 DRIVE

- Hold the tool handle (which is not intended to be weight-bearing) and start driving the SIDELoc™ Screw on the outside edge of the board running the drill at full speed. Let the screw do the work—forcing the screw can result in mushrooming or splitting. Once you've started the screw, gently increase pressure.
- When the shoulder of the driver bit reaches the screw guide, you are finished driving the screw.
- Follow the same steps to fasten the inside edge of the board.
- Continue this process down the length of the board until you've installed the entire board. Do not skip any joists. Two screws are required at every joist location, as well as 2 screws within ½" of each board end.

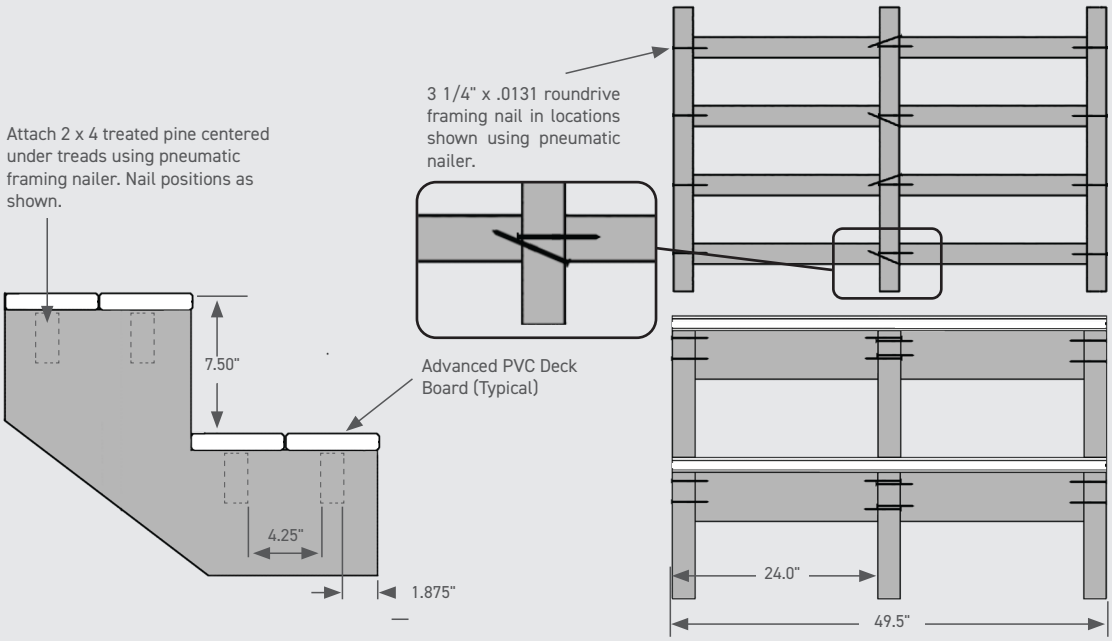


The maximum allowable stringer spacing is as shown:

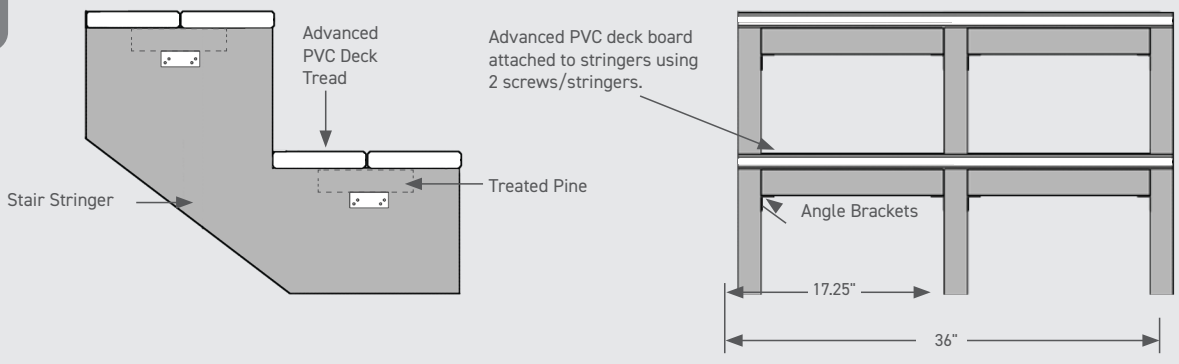
1



2



3



- Stairs should be constructed per local code requirements.
- For riser installation, you can use the Advanced PVC Riser Board or you can use a ripped piece of Advanced PVC Fascia using the fascia install recommendations on the following page.
- Alternatively, use an Advanced PVC 7.25" wide-width deck board. Attach using (2) screws every 12" maximum.
- Only use top-down fastening methods such as Cortex® or TOPLoc® stainless steel for Advanced PVC with square-shoulder deck boards for stairs.

Fastening Methods: Advanced PVC Fascia Installation Guidelines

Recommended Fasteners

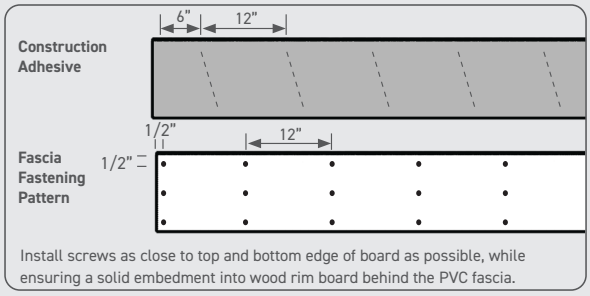
- Due to the durability of TimberTech Advanced PVC Decking products, use Advanced PVC TOPLoc® color-matched fascia fasteners for Advanced PVC Fascia, or a high-quality stainless steel screw (minimum #7 size and minimum length of 1-5/8").
- Avoid using flathead screws. Trim head screws typically provide a better result. As always, **the fastener should be tested on a sample board before being used on decking.**
- For salt water coastal applications, we suggest using the above minimum fastener requirements in 316 stainless steel.
- **Do not use Cortex® fastening system for Advanced PVC Fascia.** TimberTech Advanced PVC Fascia installation video https://www.timbertech.com/resources/installation-guides/

Storage

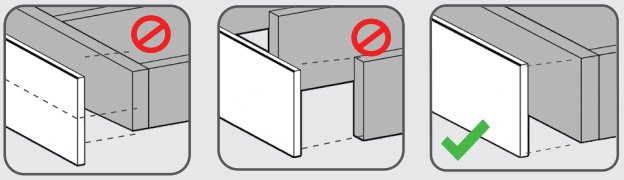
- Store fascia boards in a cooler, shady area prior to installation.
- Store fascia boards on a flat solid surface.

Fascia

- Be sure to keep butt joints tight. Do not leave a gap at butt joints, splices or miters.
- DO NOT scarf cut at butt-joints. Butt-joint should be 90° cuts.
- Keep Advanced PVC Fascia as cool as possible during attachment.
- Attach fascia to a solid backer wide enough to support the entire width of the rim cover. Use 2 x 12 wood rim board or rip cut fascia to fit. A slight overhang of 1/2" or less is okay. Do not overhang fascia more than 1/2" over the width of the wood backer.
- Use the 1 5/8" color match stainless steel TimberTech Fascia screws for PVC. DO NOT use composite fascia screws.
- Never span Advanced PVC Fascia across joist ends or hollow cavities. Always attach to a solid backer.
- Use a quality exterior-rated construction adhesive that is compatible with Advanced PVC.
- Always trim/cut the factory cut ends of fascia to ensure they are square and for a finished look.
- Advanced PVC Fascia boards are one-sided. Fascias are to be installed grain side out and always attached to a solid backer.
- Three (3) PVC TOPLoc® Fascia screws are recommended for Advanced PVC Fascia. Install in rows from top to bottom with top and bottom screws positioned no more than 1/2" from the edge as shown below. The third screw should be approximately centered in the board. Repeat this configuration every 1/2". Screws at ends of boards should be no more than 1/2" from the end at butt joints, and should be as close to end of board as possible at corners/overlaps making sure that screws insert into solid wood rim board.



* If using fascia to cover the edge of your deck boards, lower the top fasteners to 1 1/2" from the edge

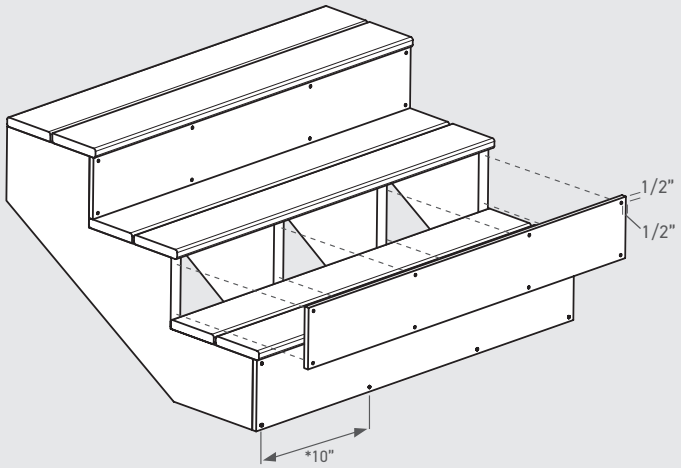


Incorrect: Overhang | Incorrect: Missing solid backer | Correct

Riser

- Be sure to keep butt joints tight. Do not leave a gap at butt joints, splices or miters.
- DO NOT scarf cut at butt-joints.
- Keep Advanced PVC Riser boards as cool as possible during attachment.
- Cortex®, TOPLoc®, and TOPLoc® for PVC fascia are all fastening options recommended for TimberTech Advanced PVC Riser boards.
- Always trim/cut the factory cut ends of riser to ensure they are square and for a finished look.
- Advanced PVC Riser boards are one-sided. Risers are to be installed grain side out.
- Install two (2) screws from top to bottom with top and bottom screws positioned no more than 1/2" from the edge as shown below. Repeat this configuration on every stringer. Unlike fascia, riser boards do not require a solid backer. Reference note below.

Note: 2 screws must be applied to each stringer. If the Riser board spans longer the 8 ft use 3 screws per stringer.



*For stringers spaced greater than 10"oc, solid blocking (min. thickness of 1.5") must be applied behind the riser board with riser fasteners every 12" or less. (No adhesive required)

The techniques shown above should be used for best results. Results may vary as expansion and contraction could still occur. TimberTech claims no liability or responsibility for the improper installation of this product. Since all installations are unique, it is the sole responsibility of the installer to determine specific requirements in regard to each decking application. TimberTech recommends that all designs be reviewed by a licensed architect, engineer or local building official before installation. Please contact TimberTech Customer Service prior to installing if you have question or concerns.

Storage & Handling

- Store TimberTech Advanced PVC Deck, Fascia, and Riser boards on a flat and level surface.
- Store products in a cool, shady area prior to installation.
- Do not stack bundles over 8 units high.
- Always leave factory-applied protective wrap. If stored outdoors, the product must be covered with non-translucent material.
- Use care when handling product with a forklift, as forks can easily damage the material.
- If banding is used, use protective corners to prevent indents from the bands.
- Avoid storing TimberTech Advanced PVC Deck, Fascia and Risers in areas of excessive heat.

Limited Warranties

TimberTech Products are made exclusively from technologically advanced materials designed to provide years of enjoyment. TimberTech Advanced PVC Decking products are covered by a Limited Lifetime Warranty for residential applications and a 20-Year Limited Warranty for commercial applications. The warranty covers defects resulting in blistering, peeling, flaking, cracking, splitting, cupping, rotting or structural defects from termites or fungal decay. TimberTech Advanced PVC Decking products are also covered by a 50-Year Fade & Stain Limited Warranty for residential applications. Visit TimberTech.com to view complete warranty information.

Care & Maintenance

To keep TimberTech Advanced PVC Deck, Fascia, and Riser looking its best:

- To remove rust stains, use a cleaner such as Krud Kutter "the must for rust".
- Just like with hardwood flooring, do not use non-porous items such as rubber or vinyl products (i.e. welcome mats, planters, pool toys, tarps, etc.) on deck boards. A reaction can occur that causes discoloration of the decking under the rubber/vinyl product. This is a common caution for polymer decking products.
- Some products, such as sunblock and insect repellent, contain chemicals that may alter the surface of Advanced PVC Deck, Fascia, and Risers. Check product labels and consult with the manufacturer as to product compatibility with plastic materials, such as Advanced PVC Deck, Fascia, and Risers, prior to use on or near Advanced PVC Deck, Fascia, and Risers.
- Do not use any PVC glue/ cement or similar product for install, butt joints, miter etc. If it gets on the surface of the board, it may discolor and permanently damage the surface.
- For light cleaning of Advanced PVC Deck, Fascia, and Risers, you can use clean, cool water to rinse most dirt or debris from the surface of the deck. For more moderate or heavy cleaning, we recommend using TimberTech DeckCleaner™ with a stiff natural fiber brush. Start by cleaning a small section of the deck prior to cleaning the entire deck to verify effectiveness. Keep in mind that in highly soiled/stained areas multiple cleanings may be needed for best results. Be sure to thoroughly rinse deck immediately after cleaning.
- Always read the cleaning product manufacturers specific information before using any product on Advanced PVC Deck and follow their instructions. Outside of the list above, it is also a good idea to test the cleaner on a scrap piece or inconspicuous area of the deck to make sure it does not harm the surface.
- WARNING: Keep children and pets away from cleaning products and Advanced PVC Deck until dry.
- *Note that composite deck cleaners such as Cortex® Clean®, Thompson's® Water Seal® Oxy Action, Olympic® Premium Deck Cleaner, or other cleaners specified as composite deck cleaners, in powder or liquid form, SHOULD NOT be used with TimberTech Advanced PVC Deck, Fascia, and Riser. Advanced PVC is a PVC/ polymer deck plank and not a wood/ plastic composite.
- Note: If any of the suggested cleaning methods are not satisfactory, take before and after photos of the attempt and contact TimberTech Customer Service at 1(877) 275-2935.
- It is always a good idea to test on a scrap piece or confirm cleaning process prior to cleaning the entire deck or large area.
- Store products under cover to maintain a clean surface.
- If stored outdoors, they must be covered with a tradnslucent material.
- **Do not use any other cleaners, detergents, stain removers, or chemicals including but not limited to: Goof Off products, Acetone, Goo Gone, Mineral Spirits, etc.** As noted, contact TimberTech Customer Service with before and after photos if any suggested methods are not satisfactory for possible further recommendations at 1(877) 275-2935.

Ice and Snow:

- As with any walking surface, Advanced PVC deck products may become slippery in Winter weather. Take caution when walking on the your Advanced PVC deck in these conditions.
- For ice removal, most products containing calcium chloride in flake or pellet form can be used without damage to the deck surface. For more pet friendly options, most products containing magnesium chloride can also be used without damaging the deck surface.
- These products may leave a white residue which can be cleaned using the cleaning guidelines above.
- Metal shovels or plastic shovels with a metal leading edge will damage the surface of Advanced PVC Decking and are not recommended for clearing snow from deck surface.

For specific details, see TimberTech Care & Cleaning Guide. https://www.timbertech.com/resources/care-cleaning/

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